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## 📊 PROJECT INSIGHTS

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### ◈ 1. Data Quality Insights

- The dataset initially contained **missing values** in key features like:
  - annual_income
  - credit_score
  - employment_type
- Different imputation techniques were applied:
  - Simple methods (mean/mode) were fast but less accurate
  - **KNN and MICE provided more realistic values**

**👉 Insight:**

Advanced imputation techniques preserve relationships between variables better than simple methods.

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### ◈ 2. Missing Data Pattern

- Missing values were **not completely random**
- Financial variables had higher missing rates

**👉 Insight:**

Missingness in financial data may indicate risk or incomplete records, which can itself be useful information.

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### ◈ 3. Outlier Behavior

- Significant outliers observed in:
  - annual_income
  - loan_amount
  - credit_score
- Removing outliers reduced dataset size
- **Winsorization preserved data while controlling extremes**

**👉 Insight:**

In financial datasets, extreme values are common and should be treated carefully rather than removed.

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### ◈ 4. Feature Distribution

- Many numerical variables were **skewed**
- After applying transformations:
  - Distribution became more normal

### 👉 Insight:

Transformations like log and Yeo-Johnson improve model performance by stabilizing variance.

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### ◈ 5. Encoding Impact

- Categorical variables were handled differently:
  - Ordinal → meaningful order preserved
  - One-hot → no false relationships

### 👉 Insight:

Choosing the correct encoding method prevents introducing bias into the model.

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### ◈ 6. Feature Engineering Insights

New features created:

**✓ Debt-to-Income Ratio**

- Strong indicator of financial risk

**✓ Average Monthly Transactions**

- Reflects user activity

**✓ Spending-to-Income Ratio**

- Shows financial discipline

### 👉 Insight:

Engineered features capture real-world financial behavior better than raw data.

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### ◈ 7. Scaling Observations

- Different scaling methods tested:
  - StandardScaler
  - MinMaxScaler
  - RobustScaler
- **RobustScaler performed best** due to outliers

👉 **Insight:**

Choice of scaling method significantly affects model stability.

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### ◈ 8. Data Transformation Insights

- Log and power transformations reduced skewness
- Improved numerical stability

👉 **Insight:**

Proper transformations help algorithms converge faster and perform better.

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### ◈ 9. Overall Data Readiness

After preprocessing:

- No missing values ✓
- Outliers controlled ✓
- Features encoded ✓
- Data scaled ✓

👉 **Insight:**

The dataset is now structured and optimized for machine learning models.

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### ◈ 10. Business Insight (VERY IMPORTANT 🔥)

From the data:

- Customers with:
    - High **debt-to-income ratio**
    - Low **credit score**
    - High **missed payments**

👉 Are more likely to default

👉 **Insight:**

Financial behavior indicators are stronger predictors than demographic data.

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### ◈ 11. Model Readiness Insight

- Dataset now supports:
    - Classification models (Logistic Regression, Random Forest, etc.)

👉 **Insight:**

Proper preprocessing is more important than complex models for performance.

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“The project demonstrates that careful preprocessing, feature engineering, and data transformation significantly enhance the quality and predictive power of the dataset, making it suitable for robust machine learning applications.”

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